

MELLE, Belgium

WMO: 064340

Lat: 50.9803N

Lon: 3.8158E

Elev: 15

StdP: 101.14

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.0	-3.6	-9.9	1.6	-4.7	-6.9	2.1	-2.1	11.4	10.5	10.7	9.9	2.9	60	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	29.5	20.2	27.1	19.1	25.2	18.0	21.1	28.0	20.0	25.8	19.0	23.5	3.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.6	23.3	17.9	12.9	22.2	17.1	12.2	21.4	60.8	28.0	57.3	25.9	54.0	23.5	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.4	8.2	7.2	-8.4	34.1	4.1	2.3	-11.4	35.8	-13.7	37.2	-16.0	38.5	-19.0	40.2
			-8.8	23.3	4.0	0.7	-11.7	23.8	-14.0	24.2	-16.3	24.5	-19.2	25.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	3.8	4.2	7.1	10.3	13.5	16.7	18.6	17.9	15.2	12.3	8.0	5.2	
	DBStd	6.20	4.05	3.98	3.48	3.56	3.59	2.97	2.98	2.51	2.65	3.49	3.53	3.99	
	HDD10.0	752	192	162	100	39	7	0	0	0	1	17	79	154	
	HDD18.3	2734	449	394	348	241	157	67	32	37	100	189	310	408	
	CDD10.0	1158	1	1	10	48	114	202	267	246	157	87	19	4	
	CDD18.3	98	0	0	0	0	6	19	41	25	7	1	0	0	
	CDH23.3	912	0	0	0	14	64	177	376	222	53	6	0	0	
	CDH26.7	243	0	0	0	0	7	43	116	67	10	0	0	0	
Wind	WSAvg	3.5	4.0	4.1	3.8	3.4	3.4	3.3	3.0	2.9	2.9	3.3	3.7	4.3	
Precipitation	PrecAvg	823	70	62	53	48	60	74	78	86	64	69	78	81	
	PrecMax	1011	161	134	131	108	119	150	156	187	181	128	134	168	
	PrecMin	632	4	17	3	0	15	27	22	20	14	14	11	22	
	PrecStd	94	35	27	30	27	25	30	36	50	37	26	30	40	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.8	19.4	25.2	27.6	30.8	33.7	32.0	28.0	23.6	17.5	14.0	
		MCWB	11.6	9.4	12.2	15.0	18.3	21.2	21.1	21.3	19.9	18.1	14.9	11.8	
	2%	DB	11.8	11.9	16.8	22.7	25.2	27.6	29.7	28.4	24.6	21.1	15.9	12.8	
		MCWB	10.3	9.6	11.6	14.4	16.8	19.4	20.4	20.1	18.2	16.9	13.6	11.3	
	5%	DB	10.4	10.7	14.7	19.8	22.9	25.2	27.1	25.6	22.0	18.6	14.5	11.8	
		MCWB	9.1	8.8	10.5	13.2	15.9	18.1	19.1	18.7	17.1	15.3	12.5	10.4	
	10%	DB	9.1	9.6	12.6	16.7	20.2	23.0	25.0	23.3	20.2	17.2	13.1	10.5	
		MCWB	8.0	8.0	9.6	11.7	14.7	17.2	18.3	17.5	16.0	14.6	11.4	9.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	10.8	13.7	16.5	20.3	21.9	22.1	22.6	20.7	18.4	15.9	12.5
			MCDB	12.7	13.0	18.0	23.4	25.9	28.4	30.0	29.2	26.9	23.0	17.2	13.3
2%		WB	10.4	9.9	12.1	14.7	18.4	20.3	21.0	20.6	19.1	17.0	13.9	11.5	
		MCDB	11.4	11.4	15.6	21.0	22.2	26.0	28.6	27.1	22.9	20.9	15.5	12.6	
5%		WB	9.1	9.1	11.0	13.4	16.5	18.9	20.0	19.3	17.7	15.9	12.7	10.6	
		MCDB	10.2	10.6	14.2	18.8	21.4	23.9	26.0	24.2	20.8	18.3	14.4	11.7	
10%		WB	8.1	8.1	9.9	12.3	15.2	17.6	18.9	18.3	16.7	14.9	11.5	9.4	
		MCDB	9.1	9.6	12.2	16.4	19.8	21.9	23.3	22.2	19.6	16.9	13.0	10.4	
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.5	7.9	9.8	9.4	9.2	9.8	9.5	9.0	7.0	5.2	4.4	
		MCDBR	5.0	7.1	12.1	14.5	13.4	13.5	13.9	14.1	11.6	9.1	6.0	4.6	
		MCWBR	4.7	5.2	7.6	7.6	6.6	6.6	6.3	7.0	6.4	5.5	4.8	4.0	
	5% WB	MCDBR	4.9	6.6	10.5	12.5	11.1	11.8	11.9	12.3	10.3	8.4	5.5	4.7	
		MCWBR	4.7	5.2	7.1	6.8	6.1	6.2	6.0	6.6	6.1	5.2	4.7	4.3	
Clear-Sky Solar Irradiance	taub	0.333	0.354	0.389	0.418	0.429	0.425	0.428	0.418	0.393	0.369	0.345	0.329		
	taud	2.371	2.339	2.238	2.184	2.202	2.233	2.230	2.276	2.335	2.392	2.375	2.341		
	Ebn at Noon	680	765	799	819	828	833	823	811	790	737	656	616		
	Edh at Noon	64	85	112	132	136	133	131	120	101	80	64	57		
All-Sky Solar Radiation	RadAvg	0.78	1.47	2.67	4.29	5.05	5.49	5.19	4.42	3.31	1.92	0.92	0.61		
	RadStd	0.08	0.28	0.35	0.44	0.42	0.46	0.56	0.40	0.26	0.16	0.10	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (72 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page